

컴퓨터 비전 개론(공) (CSE404-01)

학기	2026년 1학기
담당교수	임성훈
이수구분	교선평수
학점	3.0
강의시간	화09:00-10:30(E7 - L21), 목09:00-10:30(E7 - L21)
강의실	
소속	기초학부

교과개요

This course provides a comprehensive introduction to computer vision, covering both the theoretical foundations and practical algorithms for analyzing and understanding visual data. Topics range from image formation and processing to geometry-based vision, 3D reconstruction, and modern learning-based methods for recognition and scene understanding.

교과목표

By the end of this course, students will understand core principles of image formation, camera geometry, and multi-view vision, as well as key algorithms for feature extraction, matching, and visual perception. The course aims to equip students with the conceptual tools needed to analyze vision problems and prepare for advanced study or research in computer vision.

주차별 강의계획

1주차:Course introduction
2주차:Image formation
3주차:Image processing
4주차:Feature detection and matching
5주차:Camera geometry
6주차:Two-view geometry
7주차:3D reconstruction
8주차:Mid-term exam
9주차:Computational imaging
10주차:Image classification
11주차:Object detection
12주차:Semantic segmentation
13주차:Optical Flow
14주차:Alignment and Tracking
15주차:Computational imaging
16주차:Final exam

교과정책(방침)

This class is recommended for students who are familiar with Python programming and general mathematics (linear algebra, calculus)